

Decision and Risk Analysis

Introduction

DAEN 427/ISEN 427/ISEN 627

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What have I gotten myself into?

DAEN/ISEN 427/627 is a rigorous introduction to decision and risk analysis. The course:

- **Introduces core Bayesian statistical methods** used in data science and engineering research;
- **Emphasizes understanding**, data-driven decision-making, clear communication of results, and critical evaluation of evidence;
- **Draws on relevant examples** from engineering, psychology, economics, and public health policy; and
- **Uses modern software** to reproducibly analyze and interpret data in support of sound scientific conclusions.

When flipping a fair coin, we say
"the probability of flipping Heads is 0.5."

How do you interpret this probability?

- If I flip this coin over and over, roughly 50% will be Heads.
- Heads and Tails are equally plausible.
- Both a and b make sense.

Decision and Risk Analysis

Good decisions require four questions:

Choices → Outcomes → Uncertainty → Preferences

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1. **What can we do?**

Identify our possible choices.

2. **What could happen?**

Think about possible outcomes and probabilities.

3. **What if we are unsure?**

Make decisions even when information is incomplete.

4. **What matters most?**

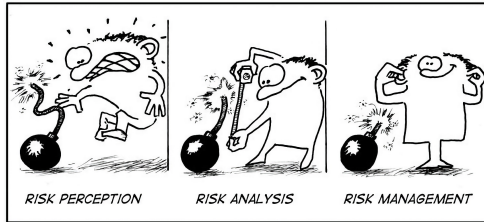
Compare outcomes, trade-offs, and preferences.

500 kg of US dollars or 500 kg of gold.

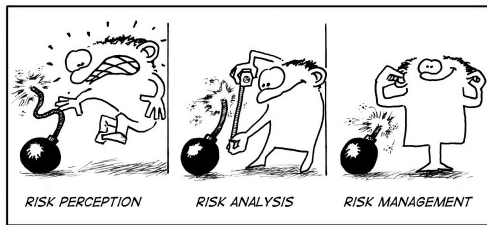


What would you choose? What would someone else choose?

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or merely the aspects of uncertainty that are easiest to quantify?



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Example: A flood risk model might estimate: "There is a 2% annual probability of flooding, with an expected property loss of \$40,000." That is mathematically convenient: *probability x expected loss*. But it may ignore whether residents trust the model or can afford insurance, or political leaders delay action, or infrastructure fails under stress. So the question becomes: **"Are we modeling the full risk of flooding, or just the measurable part: water levels, probabilities, and dollar losses?"**

Alexander Abuabara, PhD

- About me: <https://abuabara.github.io>
- **To be a great teacher ...**



- **To be a great teacher ...** that means always learning, adapting, and getting better each and every day.
- **That you learn** concepts, analyze data, pass the class ...
... and go do great things in life!



*Yo, I'll tell you what I want, what I really really want**

* Wannabe by Spice Girls

- At first, it may seem that answers are the most important.
- But the real goal is **to learn how to ASK questions**.
- When you can create thoughtful questions about a topic means you truly understand it.



Next: **Syllabus!**